

AURA Forge ADR Package

Current evidence packet · 9 tracked Markdown source(s) · 2026-09-01

ADR 0001: Quality Rubric And Evaluation Design

Source: docs/adr/0001-quality-rubric-and-evaluation-design.md

Context

The capstone needed comparable evidence across PRE-AURA control runs and AURA adaptive-route runs. The measured comparison used docs/capstone/quality-rubric.md, control evidence committed through 75dd1f8, and AURA evidence committed through 6b52ecb.

Decision

Use a 16-point rubric with four dimensions: Outcome Accuracy, Route Appropriateness, Governance & Approval Fidelity, and Evidence & Readiness Quality.

Rejected Alternatives

- Single pass/fail result only. Rejected because LOW control produced correct output but was over-served; this needed a route-quality dimension.
- Cost-only comparison. Rejected because HIGH intentionally preserved governance and should not be optimized only for cost.

Evidence

- PRE-AURA LOW: 14/16 FAIL, cost \$0.6066006, route appropriateness 2/4.
- AURA LOW: 16/16 PASS, cost \$0.3377550.
- AURA aggregate: quality improved from 44/48 to 48/48.

Consequences

The rubric supports measured route-quality claims without projecting company-wide savings.

Open Risks

The rubric is calibrated to three representative scenarios, not every possible future HIGH production change.

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ADR 0002: Memory, Context, And Prompt Architecture

Source: docs/adr/0002-memory-context-and-prompt-architecture.md

Context

The project uses role prompts, bounded evidence packets, and local memory rules. Historical calibration evidence in docs/calibration-log.md showed output-contract failures and later correction. The backend config auditor also preserved evidence of source precision problems in docs/agents/backend-config-docs-auditor/lab/iteration-log.md.

Decision

Keep role prompts narrow, pass only required evidence, and preserve measured role outputs as artifacts. Treat memory as context support, not as proof; current repository files and run artifacts remain authoritative.

Rejected Alternatives

- Broad shared context for every role. Rejected because it increases leakage and stale-evidence risk.
- Memory-only evidence. Rejected because memory may be stale and cannot replace committed files or measured artifacts.

Evidence

- Calibration before-fix run CAL-20260802-002 preserved a Tester output-contract failure.
- AURA HIGH used separate Planner, Implementer, Reviewer, Tester, and Project Manager evidence with zero authorization denials.

Consequences

Roles stay easier to audit and route metrics remain comparable.

Open Risks

Long multi-role runs still require careful artifact naming and final sanity checks to avoid mixing stale attempts with valid attempts.

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ADR 0003: MCP Tool Boundaries And Least Privilege

Source: docs/adr/0003-mcp-tool-boundaries-and-least-privilege.md

Context

Module 4 governance depends on role-scoped MCP tool grants. The sandbox foundation verified governed MCP storage/retrieval imports and read-only workspace behavior. The fresh overreach demo GO-20260811-001 proved the implementer role cannot use Project-Manager-only task_tracker.

Decision

Keep role/tool boundaries explicit and testable. Use real authorization denials as governance evidence.

Rejected Alternatives

- Trusting prompt instructions alone. Rejected because prompt-only boundaries are not enforceable.
- Granting broad shell or task-tracker access to implementation roles. Rejected because implementation should not mark work complete.

Evidence

- eval/test_policy.py, eval/test_mcp_runtime.py, and eval/test_coursetools_runtime.py are permanent deterministic checks.
- docs/capstone/governance-overreach-demo.md records a real denial: implementer was not on the allow-list for task_tracker.

Consequences

The system can show least-privilege enforcement with concrete machine evidence.

Open Risks

The course task_tracker is simulated. It proves local role authorization, not behavior of a real external ticketing integration.

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ADR 0004: Adaptive Risk Routing

Source: docs/adr/0004-adaptive-risk-routing.md

Context

The PRE-AURA route sent LOW, MEDIUM, and HIGH through the same full route. AURA introduced deterministic aura-risk-v1 and aura-router-v1, frozen at c844db1b457712d4c68c9353c49e8bd9fd2121a1.

Decision

Use deterministic path/metadata classification and fixed route plans per tier. Do not tune the router during measurement.

Rejected Alternatives

- Model-decided risk routing. Rejected because routing must be reproducible and auditable.
- Always using the full route. Rejected because LOW measured as materially over-served.
- Always minimizing cost. Rejected because HIGH must preserve strong governance.

Evidence

- LOW: model roles reduced from 5 to 2, human checkpoints from 2 to 0, quality improved from 14/16 to 16/16.
- MEDIUM: model roles reduced from 5 to 3, human checkpoints from 2 to 1.
- HIGH: retained 5 model roles and 2 human checkpoints.

Consequences

The route is risk-sensitive without being ad hoc.

Open Risks

Future HIGH categories such as deployment, auth, database migration, or production configuration may need extra deterministic gates beyond the measured scenario.

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ADR 0005: Governance And Eval-Gated CI

Source: docs/adr/0005-governance-and-eval-gated-ci.md

Context

Before this milestone, `.github/workflows/test.yml` contained ordinary backend/frontend checks but not the official Module 4 governance pattern. Commit `a102a79` added `.github/workflows/ci.yml`, `eval/ci_change_classifier.py`, and pipeline integrity checks.

Decision

Add a separate capstone CI workflow with jobs for change classification, permanent policy tests, conditional evaluation, pipeline integrity, advisory review, and audit trail.

Rejected Alternatives

- Replacing `test.yml`. Rejected because the existing application test workflow remains useful and did not need deletion.
- Making AI review blocking. Rejected because the official lesson treats it as advisory and secrets may be unavailable.
- Skipping audit on failure. Rejected because audit evidence is most useful when prior jobs fail.

Evidence

- Local workflow YAML parsing passed.
- `scripts/check-pipeline-integrity.py` rejects policy continue-on-error, missing audit, broad write permissions, production markers, and unsafe secret exposure.
- Docker policy/MCP/coursetools checks reported 18 passed.

Consequences

Permanent deterministic gates can fail CI without requiring AI secrets. Advisory review degrades to SKIPPED -- AI SECRET UNAVAILABLE.

Open Risks

Actual GitHub Actions execution is not proven until the branch is pushed with Muhammad's explicit authorization.

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ADR 0006: Agent-To-Deterministic Conversion

Source: docs/adr/0006-agent-to-deterministic-conversion.md

Context

The backend-config-docs-auditor performed useful source comparison work but historical Run 5 still had factual precision weaknesses. Commit 9f9e461 converted one stable factual subtask to deterministic code.

Decision

Convert only the DATABASE_URL representation check across README, env template, implementation, and focused tests. Leave prioritization and prose recommendation work agentic.

Rejected Alternatives

- Converting the whole auditor. Rejected because broader audits require judgment.
- Calling the AURA risk classifier the conversion. Rejected because it was designed deterministic from the start.

Evidence

- docs/agents/backend-config-docs-auditor/lab/iteration-log.md records Run 5 inspecting README, .env.example, implementation, and tests for DATABASE_URL.
- scripts/check-config-docs-consistency.py now emits deterministic JSON with model cost \$0.
- eval/test_config_docs_consistency.py covers pass and fail behavior.

Consequences

Repeated factual guardrails no longer require a model call.

Open Risks

The utility covers one stable fact only. It must not be presented as a complete backend configuration audit.

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ADR 0007: Change Passport Evidence Design

Source: docs/adr/0007-change-passport-evidence-design.md

Context

The capstone produced real classifier, router, role, test, approval, and metrics artifacts. A terminal summary alone was not enough to make the evidence portable and machine-checkable.

Decision

Implement Change Passport as an evidence aggregator. A field exists only when automatically derived from machine evidence or an actual human-approval record.

Rejected Alternatives

- Manually filled passport fields. Rejected because they would create unverifiable claims.
- Adding CI fields before GitHub Actions runs. Rejected because local CI structure is not real GitHub status evidence.
- Making Passport a new router. Rejected because classification and routing already have deterministic producers.

Evidence

- Commit 13eda9c added scripts/build-change-passport.py and tests.
- Generated local passports for LOW, MEDIUM, and HIGH read final metrics, quality scores, classifier/router evidence, approval artifacts where present, and evidence hashes.

Consequences

Passport output is deterministic and traceable to files.

Open Risks

CI fields must be added only after real GitHub Actions producers exist.

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ADR 0008: Governed Semantic Retrieval

Source: docs/adr/0008-governed-semantic-retrieval.md

Context

The existing retrieval server enforced role classification ceilings but used exact substring matching and returned no source citation. Canvas feedback specifically identified semantic document search as missing.

Decision

Keep retrieval local and deterministic for the course corpus. Convert query and document text to expanded token vectors, rank with cosine similarity, enforce the existing role ceiling before returning text, and attach a repository path and section citation to every match.

Rejected Alternatives

- External embedding API. Rejected because offline grading should require no secret, network, paid call, or production data.
- Exact substring search only. Rejected because it cannot satisfy related-term queries such as package audit against dependency/manifest/report language.
- Returning uncited snippets. Rejected because a grader or reviewer must be able to inspect the stated source.

Evidence

- eval/test_retrieval_behavior.py covers related-term ranking, citations, classification withholding, and empty-query denial.
- docs/governance-policy.md records the schema, classification field, citation fields, and bounded top_k range.

Consequences

The result is reproducible and governable without external infrastructure. Search breadth is intentionally limited by the small synthetic corpus and checked synonym map.

Open Risks

This is not a production embedding index and does not establish retrieval quality at real FitGPT document scale.

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ADR 0009: Offline Verification And Execution Limits

Source: docs/adr/0009-offline-verification-and-execution-limits.md

Context

Canvas feedback asked for stronger network/parallel-session boundaries and demonstrated timeout, retry, and budget enforcement. Historical AURA measurements and holdouts must remain unchanged.

Decision

Add two independent deterministic controls:

- an offline Docker verifier with network disabled, read-only root/workspace, no credentials, dropped capabilities, bounded resources, tmpfs state, and unique container naming;
- an execution-limit decision function that evaluates recorded attempt count, elapsed time, cost, and failure class.

Rejected Alternatives

- Rerunning paid historical workflows. Rejected because it would spend model budget and change the evidence envelope without user approval.
- Modifying frozen aura-router-v1. Rejected because the measured route must remain frozen.
- Document-only reliability claims. Rejected because the grader requested demonstrated enforcement.

Evidence

- eval/test_sandbox_contract.py checks the container boundary contract.
- eval/test_reliability_controls.py covers retry allowance, retry exhaustion, timeout, budget stop, and escalation.
- .github/workflows/ci.yml includes both test families in permanent gates.

Consequences

Graders can verify deterministic governance without model credentials or egress. Attempt continuation decisions become reproducible and fail closed.

Open Risks

Docker Desktop must be running for the offline command. The fresh offline run passed 32 tests. Historical model-backed runs used bridge networking and remain labeled with that limitation.